

Gabriel Schütz de Souza

Senior Software Engineer — Airport Revenue & Payment Systems

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English (fluent) · Portuguese (native) · German (fluent, TestDaF 5/5/5/4) · French (B2)

Six years building and running **approach-fee billing, collection, and payment infrastructure** in production for a German aviation SaaS serving 350+ airfields: fee calculation from flight movements, customer matching, invoice generation, overdue collection, and end-to-end card/SEPA payment pipelines. The systems where a mistake costs money or breaks the law. The billing engine I bootstrapped has **invoiced and collected €62.9M in approach fees**.

MOST RELEVANT EXPERIENCE — AIRPORT REVENUE & BILLING

Senior Software Engineer — Payments & Financial Systems

aerops GmbH (German aviation SaaS, 350+ airfields) · Jul 2020 – Jul 2026 · Remote contract (via own consultancy)

Approach-fee billing & collection:

- Built **AFS, the approach-fee billing service** for a national air-navigation services provider, in Node.js (v1 solo, 2021): it ingests flight movements, calculates MTOW-based **approach fees**, and issues the fee notices. It has **invoiced and collected €62.9M in approach fees** all-time, mostly bank/SEPA-settled. Peak month (July 2022): 10,856 movements and 363 fee notices.
- Ran the monthly billing rounds by hand early on, then automated the **customer matching: ~90% of movements are assigned to the right account automatically**, which is why a single operator can run the whole billing operation.
- Also worked on aerops' **airport management system** (PHP), which calculates **landing, parking, approach, and emission fees** for the airports themselves from the same flight-movement data.
- Built the **dunning / collections microservice** (sole author, from scratch) for overdue fee notices, and the per-airport **document-dispatch platform** that delivers them: SMTP and OAuth mailers with multi-tier fallback and per-mailer token-bucket rate limiting.

Payments & reconciliation:

- Re-architected the platform's **Stripe payment infrastructure**, inherited as a Shopware plugin, into a central payment service in the core backend: **Stripe Connect** destination charges with automated fee-splitting and partner payouts, webhooks, **SEPA** mandate lifecycle, 3D Secure, multi-currency (EUR/CHF), handling **€8.47M** in gross volume across the platform.
- Built the **bank-statement import + payment-matching** system (auto-reconciles by invoice number + amount): ~80% of incoming customer payments match exactly, ~97% at least get an automatic candidate. Plus a refund/storno pipeline with retry and **idempotency** handling that eliminated duplicate billing.
- Cleared request timeouts on very large orders (2,000+ line items) via a batched article-append API.

Compliance & integration (revenue you can defend in an audit):

- Built the German/EU **financial-compliance export suite** end-to-end: automated **DATEV** accounting exports, **XRechnung / EN 16931** e-invoicing built from the ground up with a validator, and a **SAP FI (RFBIBL) interface taken solo to two customer production go-lives** (with formal written customer sign-offs), plus **BMD** (Austria) and **CEGID** (France) exports.

How I work: one of three core engineers; deep, daily **AI-augmented workflow** (agentic code review, pre-deploy impact analysis, AI-driven browser testing, large cross-codebase migrations). I ship fast without giving up architectural control.

TECHNICAL STACK

Backend: Node.js/Express, PHP/Laravel · **Data:** MySQL/MariaDB (schema design, indexing), MongoDB · **Payments:** Stripe (Connect, Terminal, webhooks), SEPA · **Compliance:** DATEV, XRechnung / EN 16931, SAP FI (RFBIBL)

Messaging & async: RabbitMQ, DB-backed job queues · **Infra/CI:** Docker, GitHub Actions · **Testing:** PHPUnit, Codeception, Jest, PHPStan, Playwright · **APIs:** REST design, OpenAPI

EDUCATION & AWARDS

- **B.Sc. Computer Science — Uniritter (Brazil)**, in progress, expected 2027
- Prior CS coursework: **TU Dresden** (2019–2021); **UNISINOS** Computer Engineering (2017–2019)
- ICYS Gold Medal, Engineering — Stuttgart (2017) · FEBRACE 2nd Place — São Paulo (2017) · Weizmann Institute International Summer Science Institute scholarship (2016)